

Week 6 Project Summary

1. Planned: validate the Week 5 drivers statistically, build a combined risk-segmentation view, test the reminder effect by risk tier, and exchange refined findings with Data Science.

2. Completed: chi-square and joint logistic regression validation of all five Week 5 candidate drivers; a compound 0-3 risk score with monotonic no-show gradient; a reminder-by-risk-tier test; a cross-track integration package for Data Science.

3. Improved from Week 5: moved from purely descriptive bucketed rates to statistically validated, jointly modelled drivers, and from a single-variable risk flag (13.4% no-show coverage) to a compound risk score (62.9% coverage).

4. Integrated: refined risk segmentation and validated feature set shared with Data Science as an updated benchmark and feature-selection input.

5. Track(s) collaborated with: Data Science.

6. What was exchanged: validated features and the compound risk score outward; baseline model feature usage inward (with a placeholder for Week 6 feature-importance/error-analysis once received).

7. What changed as a result: the high-risk segment definition used across the project is now statistically grounded and captures far more of the no-show problem than the Week 5 version.

8. Key findings: all three main drivers are independent and additive, waiting time and appointment day remain non-drivers; reminders help consistently across risk tiers.

9. Major challenges: balancing risk-score sensitivity (coverage) against specificity (segment size) resolved by proposing a tiered rather than single-flag intervention.

10. Important decisions: adopted a compound (multi-variable) risk score over a single-variable rule; kept the reminder-by-risk finding flagged as suggestive rather than conclusive given borderline significance.

11. Remaining issues: risk score thresholds are heuristic, not model-optimized; awaiting Data Science's Week 6 output to complete the two-way integration loop.

12. Contribution to HealthConnect: a validated, statistically grounded patient risk-triage tool ready for operational piloting and for benchmarking the Data Science model.

13. Proposed Week 7 focus: compare the risk score against the Data Science model, design a properly powered reminder test, and re-validate findings on any updated dataset.